Unsupervised Learning for Pattern Discovery and Customer/User Segmentation

In this assignment, you will use **unsupervised learning** to discover patterns in a marketing-related dataset. Each student will receive an individual dataset and a short business plan. Although the datasets are different, all students must complete the same tasks below.

Task 1: Understand the Business Problem

Read the business plan provided with your dataset and briefly explain:

1. What is the main business problem?
   Answer: The main business problem is that the company does not fully understand the different types of customers in its customer base. As a result, marketing campaigns may be too broad and less effective, leading to lower customer engagement, reduced sales opportunities, and inefficient use of marketing resources.

2. Why is customer/user segmentation useful for this business?
   Answer: Customer segmentation is useful because it groups customers with similar characteristics, behaviors, or preferences into distinct segments. This helps the business better understand its customers, identify high-value groups, personalize marketing efforts, improve customer satisfaction, and allocate resources more effectively.

3. What kind of marketing decisions could be improved by discovering customer/user groups?

   Answer:

- Discovering customer groups can help improve several marketing decisions, including:
- Targeting the right customers with specific marketing campaigns.
- Identifying high-value customer segments for retention programs.
- Selecting the most effective communication channels for each group.
- Improving customer acquisition and loyalty strategies.

Task 2: Prepare the Dataset

Using your assigned dataset:

1. Load the dataset in Python.
2. Inspect the dataset by checking:
   - Number of rows and columns
   - Column names
   - Data types

- Missing values
- Duplicate rows

3. Clean the dataset if needed.
4. Select the numerical features that are useful for clustering.
5. Scale the selected features before applying K-means clustering.

Task 3: Apply K-means Clustering

Train a K-means clustering model using your selected features.

You must:

1. Use the **Elbow Method** to help choose a suitable number of clusters.

Answer: The Elbow Method was used by plotting inertia values for K values ranging from 1 to 10. The graph showed a noticeable bend around **K = 3**, indicating that adding more clusters beyond this point provided diminishing improvements in reducing inertia.

2. Explain why you selected that number of clusters.

Answer: A value of **K = 3** was selected because the Elbow Method showed that three clusters provided a good balance between model simplicity and cluster separation. Using more than three clusters would increase complexity while providing only small improvements in clustering performance.

3. Train the final K-means model.

Answer: The final K-Means model was trained using:

```
kmeans = KMeans(
n_clusters=3,
random_state=42,
n_init=10 )
```

The model was trained using the scaled numerical features:

- Party_Size
- Lead_Time_Days
- Prior_Reservations
- Past_No_Show_Count
- Average_Spend_Per_Visit
- Distance_From_Restaurant_KM
- Reservation_Value_Score

4. Add the cluster labels back to your dataset.

Answer: The generated cluster labels were added back to the dataset using:df_clustered["Cluster"] = cluster_labels. This allowed each customer reservation to be assigned to one of the three identified customer segments.

Task 4: Analyze and Interpret the Clusters

After creating the clusters, analyze the characteristics of each group.

You should answer:

1. How many customer/user segments did you find?

Answer: The K-Means clustering model identified **three customer segments**. These segments represent groups of customers with similar reservation and dining behaviors.

2. What are the main characteristics of each segment?

Answer:

Segment 1: High-Value Loyal Diners (Cluster 1)

- Highest Reservation Value Score (86.42)
- Highest Average Spend Per Visit ($67.08)
- Largest average party size (4.39)
- Moderate reservation history
- Low no-show behavior

These customers generate the highest value for the restaurant and consistently make valuable reservations.

Segment 2: Occasional Low-Value Diners (Cluster 0)

- Lowest Reservation Value Score (28.90)
- Lowest Average Spend Per Visit ($45.02)
- Smallest party size (1.91)
- Moderate reservation activity
- Low no-show behavior

These customers visit occasionally and contribute less revenue compared to other groups.

Segment 3: High No-Show Risk Customers (Cluster 2)

- Highest Past No-Show Count (1.64)
- Moderate spending levels
- Moderate reservation value
- Smaller party sizes
- These customers have a greater tendency to miss reservations, creating operational challenges for the restaurant.

3. How are the segments different from each other?

Answer: The segments differ primarily in their spending behavior, reservation value, party size, and no-show history.

- High-Value Loyal Diners generate the most revenue and have the highest reservation scores.
- Occasional Low-Value Diners spend less and create lower business value.
- High No-Show Risk Customers are distinguished by their significantly higher no-show rates.

4. Which segment may be the most valuable for the business? Why?

Answer: The High-Value Loyal Diners segment is the most valuable because these customers have the highest spending levels, largest party sizes, and strongest reservation value scores. Retaining and rewarding this segment can significantly increase restaurant revenue and customer loyalty.

5. Which segment may need more marketing attention? Why?

Answer: The High No-Show Risk Customers segment may require the most marketing attention. Their history of missed reservations can lead to lost revenue and unused seating capacity. Targeted reminder campaigns, deposit requirements, or loyalty incentives may help reduce no-show behavior.

Task 5: Visualize the Clusters

Create at least **two visualizations** to support your analysis.

Your visualizations may include:

- Elbow Method plot
- Scatter plot using two selected features
- Bar chart comparing cluster averages
- Box plot comparing customer/user groups
- Any other useful chart for business interpretation

For each visualization, write a short interpretation explaining what the chart shows.

Visualization 1:
Elbow Method Plot

The Elbow Method shows how inertia decreases as the number of clusters increases. The curve begins to flatten around **K = 3**, suggesting that three clusters provide a good balance between model simplicity and cluster separation. Therefore, K = 3 was selected for the final model.

Scatter Plot (Average Spend Per Visit vs Reservation Value Score)

The scatter plot shows clear separation between customer groups. Customers with higher reservation value scores generally have higher spending levels. The plot confirms that the clustering model successfully identified different customer behaviors and value levels.

Cluster Summary Bar Chart

Compares Average Spend Per Visit, Reservation Value Score and Past No-Show Count across all clusters.

Task 6: Business Interpretation

Based on your clustering results, write a short business recommendation.

Your answer should explain:

1. What patterns did you discover?

Answer: The analysis revealed three distinct customer groups based on spending habits, reservation value, and no-show behavior. Some customers consistently generate high value, while others exhibit a higher risk of missing reservations.

2. How can the business use these customer/user segments?

Answer: The restaurant can personalize marketing campaigns, loyalty programs, reservation policies, and customer engagement strategies for each customer group.

3. What marketing strategy would you suggest for each segment?

Answer:

High-Value Loyal Diners
- VIP rewards
- Exclusive promotions
- Personalized offers

Occasional Low-Value Diners
- Discounts
- Promotional bundles
- Loyalty enrollment campaigns

High No-Show Risk Customers
- Automated reminders
- Deposit requirements

- Reservation confirmation incentives

4. How can this analysis help the business improve decision-making?

Answer: The clustering analysis enables the restaurant to better understand customer behavior, allocate marketing resources effectively, reduce no-shows, improve customer retention, and increase revenue.

Task 7: Limitations and Responsible AI Reflection

Write a short reflection answering:

1. What is one limitation of your dataset or clustering model?

Answer: The dataset only contains a limited number of customer and reservation characteristics. Additional factors such as customer demographics, dining preferences, and seasonal trends could improve segmentation accuracy.

2. Why does K-means not always produce perfect customer segments?

Answer: K-Means assumes that clusters are clearly separated and roughly similar in shape. Real customer behavior is often more complex, which means some customers may not fit perfectly into a single segment.

3. What kind of bias or unfair decision could happen if the business uses these clusters without human review?

Answer: The restaurant could incorrectly assume that all customers within a cluster behave the same way. This could result in unfair treatment, inappropriate promotions, or restrictive reservation policies for some customers.

4. Why should human judgment still be used when making marketing decisions?

Answer: Human judgment is necessary to provide context, validate model results, and ensure that marketing decisions remain fair, ethical, and aligned with business goals. Clustering should support decision-making rather than replace human expertise.

End